REPORT

Lumi Threads Supply Chain Optimization Analysis

An evaluation of supplier performance, inventory management, and logistics efficiency

March 2026

COMPANY OVERVIEW

Lumi Threads is a fictional retail company that specializes in handmade crocheted clothing and accessories. The company sources yarn and textile materials from suppliers, works with crochet manufacturers and artisans to produce finished products, stores them in a central warehouse, and distributes them to retail outlets and online customers.

The company's supply chain consists of five main stages: supplier procurement, product manufacturing, warehouse storage, distribution to retail stores, and final delivery to customers. As Lumi Threads expands its product demand, efficient supply chain management becomes essential to ensure reliable supplier deliveries, optimized transportation costs, and balanced inventory levels across the network.

This project analyzes Lumi Threads' supply chain operations to identify risks such as delayed supplier deliveries, high transportation costs, and inventory shortages that may lead to stockouts. The goal is to propose strategies to optimize supply chain performance and improve operational efficiency.

PROBLEM

THE QUESTION

How can Lumi Threads optimise its supply chain to ensure prompt supplier deliveries, reduce transportation costs, support inventory levels, and minimize stockouts for crocheted clothing and accessories?

THE KEY CHALLENGES

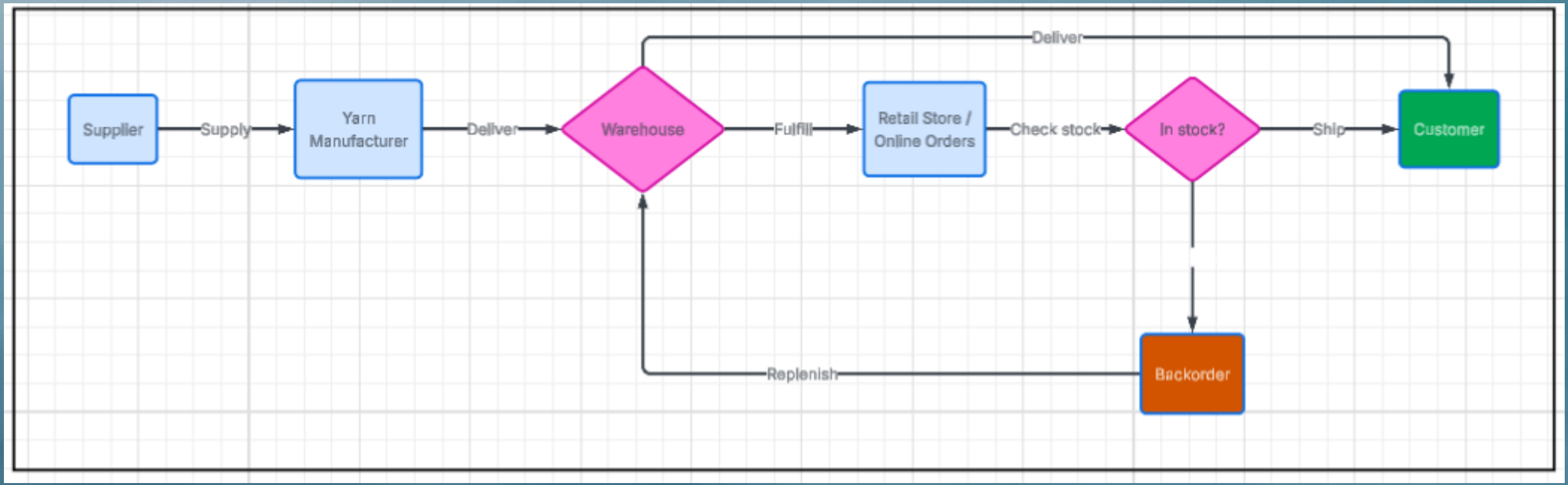
- Some yarn suppliers deliver late or unpredictably. This leads to delays in manufacturing crocheted products.
- Shipping costs vary between suppliers and fulfillment paths.
- High costs reduce profit margins for small-batch products.
- Warehouse stock sometimes falls below store demand, causing stockouts.
- Excess inventory increases storage costs and risks product aging.
- When products are not available in stores or for online orders, customers experience delays or cancellations. This impacts customer satisfaction and revenue.

PROBLEM

PROBLEM STATEMENT

Lumi Threads' current supply chain faces operational inefficiencies across supplier deliveries, coordination costs, and inventory management. These inefficiencies can lead to stockouts, delayed shipments, and increased costs. This may negatively affect customer satisfaction and profitability. The project aims to analyse these risks and propose optimisation strategies to improve overall supply chain performance.

SUPPLY CHAIN OVERVIEW



DATASET

A dataset of 40 transactions was created to simulate real-world supply chain operations. The dataset includes supplier delivery times, transportation costs, warehouse inventory levels, store demand, stockout occurrences, and fulfillment paths. The data was analysed using Power BI to generate key performance indicators and visual insights.

KEY METRICS

- Average Lead Time
- Stockout Rate
- Average Transport Cost
- Total Orders
- Inventory Gap

KEY FINDINGS

The data reveals several critical operational inefficiencies. The stockout rate is approximately 43%, indicating that a huge part of customer demand is not being met immediately. This suggests poor alignment between inventory levels and demand. A key contributor to this issue is YarnCraft Co, which shows the most negative inventory gap, highlighting its role in driving inventory shortages.

Additionally, supplier delivery times are inconsistent, reaching up to 11 days with an average of 7.38 days. This variability disrupts inventory recovery and increases the likelihood of stockouts.

Transportation costs vary significantly across suppliers, indicating inefficiencies in logistics and potential opportunities for cost optimisation. The data also shows that there is a considerable number of orders fulfilled through warehouse backorders rather than direct retail availability This results in delayed deliveries and a weaker customer experience. Overall, the negative inventory gap across the supply chain confirms a persistent imbalance between supply and demand. This reinforces the need for improved inventory planning and supplier performance management.

RECOMMENDATIONS

Lumi Threads should prioritize improving supplier selection by focusing on partners with more reliable and consistent delivery times, particularly reducing dependence on underperforming suppliers. Implementing more accurate demand forecasting techniques will help align warehouse inventory with actual store demand which will reduce both stockouts and excess inventory. Optimising transportation strategies such as consolidating shipments and selecting cost-effective transport modes can reduce logistic costs.

The business should also introduce safety stock levels to buffer against demand variability and supplier delays, to ensure better product availability.

Finally, consistent monitoring of key performance indicators using dashboards in tools such as Microsoft Power BI will allow for real-time decision-making and ongoing performance improvement throughout the supply chain.

CONCLUSION

The analysis highlights key inefficiencies within the Lumi Threads supply chain, particularly in supplier performance, inventory management, and cost control. By addressing these areas through improved supplier management, better inventory planning, and optimized logistics strategies, the business can enhance operational efficiency, reduce costs, and improve overall customer satisfaction.